Chapter 5: Nonideal Interconnect Issues

Outline

- Transmission Line Losses
- Variations in the Dielectric Constant
- Serpentine Traces
- Intersymbol Interference
- Effects of 90° Bends
- Effect of Topology

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1. Transmission Line Losses

1.1 Introduction — Why Losses Are Important

(chap5.pdf, p.3)

- Development of high-speed digital systems → requirement for smaller dimensions and faster signal rates (high-frequency content) → **resistive losses become more severe**
- Result: signal amplitude is decreased → affects noise margins, slows edge rates → affects timing margins
- Two types of dispersion:
  1. **Loss dispersion**: caused by loss variations with frequency
  2. **Phase dispersion**: caused by phase (velocity) variations with frequency

1.2 DC Losses — Conductor

(chap5.pdf, p.4)

DC conductor resistance is significant for: - Small-geometry conductors - Very long lines - Multiload buses

DC loss depends on **resistivity** and **cross-sectional area** in which current flows.

$$R = \frac{\rho L}{A} = \frac{\rho L}{Wt} \quad \text{ohms}$$

(chap5.pdf, p.4)

where ρ = resistivity, L = length, W = width, t = thickness.

- Ground return path losses are usually **negligible at DC** because the ground plane has very large cross-sectional area.

  Figure description (p.4): Cross-section of a rectangular PCB trace showing width W and thickness t. The ground plane below is shown as a thick line.

1.3 DC Losses — Dielectric

(chap5.pdf, p.5)

- Dielectric losses at DC for conventional substrates are **very negligible** and can be ignored.

> **Important note (p.5):** When frequency increases beyond DC up to 15 GHz, **the substrate (dielectric) loss dominates over the conductor loss** for FR4 substrate.

1.4 Skin Effect — Definition

(chap5.pdf, p.5–6)

At high frequencies, current flowing in a conductor migrates toward the **periphery (skin)** of the conductor.

Skin depth δ: the depth at which the electric field amplitude has decayed by a factor of e^{-1} of its surface value.

$$\delta = \sqrt{\frac{2\rho}{\omega\mu}} = \sqrt{\frac{\rho}{\pi F \mu}} \quad \text{meters}$$

(chap5.pdf, p.6)

- Approximately **63% of the total current** flows within one skin depth.

1.5 Skin Effect — Frequency-Dependent Resistance

(chap5.pdf, p.6)

As frequency increases: - Current cross-section shrinks $\rightarrow$ resistance **increases with** $\sqrt{F}$ (square root of frequency) - The transition frequency F_t is where skin depth $\delta = t$ (conductor thickness) - Below F_t: resistance $\approx$ DC value - Above F_t: resistance $\propto \sqrt{F}$

1.6 Skin Effect — Frequency-Dependent Inductance

(chap5.pdf, p.7)

As frequency increases, total inductance **falls asymptotically** toward a static value called the **external inductance** (inductance calculated assuming all current flows on the conductor exterior).

- In most high-speed digital systems, frequencies are high enough that the **frequency dependence of inductance can be ignored** (external inductance is a good approximation).

  > **Note (p.7):** Inductance is *larger* when current is distributed uniformly through the cross-section (low frequency) — analogous to even-mode excitation where mutual flux is added. At high frequencies, current concentrates on the surface, reducing internal flux linkage.

1.7 Skin Effect — Conductor Losses in a Microstrip

(chap5.pdf, p.8–12)

In a microstrip: - Current is concentrated on the **bottom edge** (pulled by fields between signal line and ground plane) - Current distribution **curves up the sides** (field concentration along thickness t)

> **Figure description (p.8):** Cross-section of microstrip trace (width W, thickness t) showing the skin depth δ on the bottom and sides. The ground plane below shows current density distribution peaking directly below the trace.

AC resistance of signal line (approximate, per unit length):

$$R_{\text{ac signal}} \approx \frac{\rho}{W\delta} = \frac{\sqrt{\rho\pi\mu F}}{W} \quad \Omega/\text{m}$$

(chap5.pdf, p.9)

Combined DC + AC resistance in SPICE:

$$R_{\text{total}} \approx \sqrt{R_{\text{ac}}^2 + R_{\text{dc}}^2}$$

(chap5.pdf, p.10)

Ground return path current density distribution (approximate):

$$I(D) \approx \frac{I_o}{\pi H} \cdot \frac{1}{1 + (D/H)^2}$$

(chap5.pdf, p.10)

where D = lateral distance from center, H = substrate height.

79.5% of ground current is within $\pm 3H$ (6H total width) $\rightarrow$ approximate ground effective width $W_{\text{eff}} = 6H$:

$$\int_{-3H}^{3H} \frac{I_o}{\pi H} \frac{1}{1 + (D/H)^2} = \frac{2I_o}{\pi} \tan^{-1}(3) \approx 0.795 I_o$$

(chap5.pdf, p.11)

Ground return AC resistance:

$$R_{\text{ac ground}} \approx \frac{\rho}{A_{\text{ground}}} = \frac{\rho}{6\delta H} = \frac{\rho}{6H}\sqrt{\frac{\pi\mu F}{\rho}} = \frac{\sqrt{\rho\pi\mu F}}{6H} \quad \Omega/\text{m}$$

(chap5.pdf, p.11)

Total AC resistance of microstrip:

$$R_{\text{ac microstrip}} = R_{\text{ac signal}} + R_{\text{ac ground}} \approx \sqrt{\rho\pi\mu F}\left(\frac{1}{W} + \frac{1}{6H}\right) \quad \Omega/\text{m}$$

(chap5.pdf, p.12)

More exact formula (conformal mapping):

$$R_{\text{signal}} = [\text{loss ratio}]\left(\frac{1}{\pi} + \frac{1}{\pi^2}\ln\frac{4\pi W}{t}\right)\frac{\sqrt{\pi\mu F\rho}}{W}$$

(chap5.pdf, p.12)

$$\text{Loss ratio} = \begin{cases} 0.94 + 0.132\dfrac{W}{H} - 0.0062\left(\dfrac{W}{H}\right)^2 & \text{for } 0.5 < \dfrac{W}{H} < 10 \\ 1 & \text{for } \dfrac{W}{H} \leq 0.5 \end{cases}$$

(chap5.pdf, p.12)

$$R_{\text{ground}} = \frac{W/H}{W/H + 5.8 + 0.03(H/W)} \cdot \frac{\sqrt{\pi\mu F\rho}}{W} \quad \text{for } 0.1 < \frac{W}{H} < 10$$

(chap5.pdf, p.12)

1.8 Skin Effect — Conductor Losses in a Stripline

(chap5.pdf, p.13)

In a stripline, current density depends on the proximity of **both** ground planes (H_1 above, H_2 below). The resistance is approximated as the **parallel combination** of two microstrip resistances:

$$R_{\text{ac stripline}} = \frac{R_{(H1)\text{ac microstrip}} \cdot R_{(H2)\text{ac microstrip}}}{R_{(H1)\text{ac microstrip}} + R_{(H2)\text{ac microstrip}}} \quad \Omega/\text{m}$$

(chap5.pdf, p.13)

> **Figure description (p.13):** Stripline cross-section: trace (width W, thickness t, skin depth δ) sandwiched between upper ground plane (distance H_1) and lower ground plane (distance H_2). Ground plane current density curves shown on both sides, matching the Lorentzian distribution.

1.9 Skin Effect — Surface Roughness

(chap5.pdf, p.14–15)

In reality, metal surfaces are rough (described by **tooth structure**; magnitude = **tooth size**): - When mean surface roughness is a significant percentage of skin depth → resistance **increases by 10–50%** above ideal formula values - Typical FR4 boards: average tooth size of **4 to 7 μm** - Etched surfaces: smaller tooth size

1.10 Surface Resistance (R_S)

(chap5.pdf, p.15–16)

Since $R_{\text{ac}} = R_s\sqrt{F}$, the surface resistance is:

$$R_s = \frac{L\rho}{W\delta} = \frac{L}{W}\sqrt{\rho\pi\mu} \quad \text{ohms} \cdot \sqrt{\text{seconds}}$$

(chap5.pdf, p.16)

For a square ($L = W$):

$$R_{\text{square}} = \frac{\rho L}{A} = \frac{\rho L}{tW} \quad \xrightarrow{L=W} \quad R_{\text{square}} = \frac{\rho}{t}$$

(chap5.pdf, p.16)

The AC resistance is obtained by multiplying R_S by length and $\sqrt{F}$, then dividing by width:

$$R_{\text{ac}} = R_s\sqrt{F}$$

(chap5.pdf, p.15)

1.11 Skin Effect — Effect of AC Losses on Signals

(chap5.pdf, p.17–19)

Fourier expansion of a 50% duty cycle square wave:

$$f(x) = \frac{2}{\pi}\sum_{n=1,3,5,\ldots}\frac{1}{n}\sin(2\pi nFx)$$

(chap5.pdf, p.17)

Only odd harmonics present for 50% duty cycle; even harmonics appear if duty cycle $\neq$ 50%.

Attenuation in a matched system (voltage divider at single frequency):

$$V_1 = V\frac{R_{\text{ac}} + R_2}{R_1 + R_{\text{ac}} + R_2}, \quad V_2 = V\frac{R_2}{R_1 + R_2 + R_{\text{ac}}}$$

(chap5.pdf, p.18)

$$\alpha = \frac{V_2}{V_1} = \frac{R_2}{R_{\text{ac}} + R_2}$$

(chap5.pdf, p.18)

Simulation procedure (FFT method): 1. Take FFT of the input waveform 2. Multiply each frequency component by the attenuation factor $\alpha(F)$ 3. Take IFFT to get the output waveform

Effects observed: - **Wave shape is rounded**: high-frequency components are attenuated $\rightarrow$ edge rate degrades - **Amplitude is reduced**: low-frequency (and DC) components are also attenuated

> **Figure description (p.19):** Four-panel diagram. (a) Original square wave in time domain. (b) FFT spectrum showing fundamental at 400 MHz and harmonics at 1200 MHz, 2000 MHz, etc. (c) Attenuation factor α vs. frequency (0.65–1.0), decreasing monotonically from DC toward GHz. (d) Time-domain output comparing "No AC losses" (clean square wave) vs. "With AC losses" (rounded, attenuated waveform).

1.12 Frequency-Dependent Dielectric Losses

(chap5.pdf, p.20–27)

Physics: Time-varying electric field causes **polar molecules** to align opposite the applied field (**electric polarization**). The damping mechanisms are frequency-dependent $\rightarrow$ dielectric losses.

Complex dielectric constant:

$$\varepsilon = \varepsilon' - j\varepsilon''$$

(chap5.pdf, p.20)

Loss tangent (primary characterization method):

$$\tan|\delta_d| = \frac{1}{2\rho\pi F\varepsilon} = \frac{\varepsilon''}{\varepsilon'}$$

(chap5.pdf, p.21)

This results in an equivalent **parallel conductance** per unit length:

$$G = \frac{\varepsilon''}{\varepsilon'}(2\pi F C_{11}) \quad \text{siemens/m}$$

(chap5.pdf, p.21)

where C_{11} is the self-capacitance per unit length.

> **Figure description (p.22):** FR4 loss tangent vs. frequency (0.01 MHz to 1000 MHz). The curve rises from ~0.010 at low frequencies, peaks at ~0.024 near 1–10 GHz, then decreases. The typical value used (0.02) is a reasonable approximation for frequencies in the hundreds of MHz range.

Worked example (symmetric stripline: $W = 5$ mils, $H_1 = H_2 = 10$ mils, $t = 0.63$ mil, $\varepsilon_r = 4.0$): *(chap5.pdf, p.23–27)*

Step 1 — Surface resistance and signal/ground AC resistance:

Since $W/H = 5/10 = 0.5 \rightarrow$ loss ratio $= 1.0$:

$$R_{\text{signal}} = 0.00172\sqrt{F} \quad \Omega/\text{m}$$

(chap5.pdf, p.25)

$$R_{\text{ground}} = 0.0003\sqrt{F} \quad \Omega/\text{m}$$

(chap5.pdf, p.25)

$$R_{(H1)\text{ac microstrip}} = R_{\text{signal}} + R_{\text{ground}} = 0.0020\sqrt{F} \quad \Omega/\text{m}$$

(chap5.pdf, p.25)

(Since $H_1 = H_2$, $R_{(H2)} = R_{(H1)}$)

$$R_{\text{ac stripline}} = \frac{R_{(H1)} \cdot R_{(H2)}}{R_{(H1)} + R_{(H2)}} = \frac{R_{(H1)}}{2} = 0.0010\sqrt{F} \quad \Omega/\text{m}$$

(chap5.pdf, p.25)

$$\Rightarrow R_s = 0.0010 \, \frac{\Omega}{\text{m} \cdot \sqrt{\text{Hz}}}$$

(chap5.pdf, p.25)

Step 2 — Transition frequency (where skin depth = conductor thickness):

$$\delta = \sqrt{\frac{\rho}{\pi F \mu}} \quad \Rightarrow \quad 0.63 \text{ mil} = \sqrt{\frac{1.72 \times 10^{-8}}{\pi[12.56 \times 10^{-7}]F}} \quad \Rightarrow F = 17 \text{ MHz}$$

(chap5.pdf, p.26)

- Below 17 MHz: $R_{\text{dc}} = \dfrac{\rho L}{Wt} = 8.5\ \Omega/\text{m}$
- Above 17 MHz: $R_{\text{ac}} = (0.0010)\sqrt{F} \quad \Omega/\text{m}$

Step 3 — Series resistance at 400 MHz:

$$R_{\text{ac}}(400 \text{ MHz}) = 0.0010\sqrt{400 \times 10^6} = 20.2\ \Omega/\text{m}$$

(chap5.pdf, p.27)

Step 4 — Shunt conductance (dielectric losses) at 400 MHz:

$$\text{PD} = \frac{\sqrt{\varepsilon_r}}{c} = \frac{\sqrt{4}}{3 \times 10^8} = 6.67 \text{ ns/m}$$

(chap5.pdf, p.27)

$$Z_o = \frac{60}{\sqrt{4.0}} \ln \frac{4(10 + 10 + 0.63)}{0.67\pi[0.63 + 0.8(5)]} = 64\ \Omega$$

(chap5.pdf, p.27)

$$C = \frac{\text{PD}}{Z_o} = \frac{6.67 \times 10^{-9}}{64} = 104 \text{ pF/m}$$

(chap5.pdf, p.27)

$$G = \frac{\varepsilon''}{\varepsilon'}(2\pi F C_{11}) = 0.013 \cdot 2\pi \cdot (400 \text{ MHz}) \cdot (104 \text{ pF}) = 0.0034 \text{ m}/\Omega$$

(chap5.pdf, p.27)

$$\Rightarrow \frac{1}{G} = 294\ \Omega/\text{m}$$

(chap5.pdf, p.27)

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2. Variations in the Dielectric Constant

2.1 PCB General Properties

(chap5.pdf, p.28)

ε_r affects: propagation velocity, characteristic impedance, and crosstalk.

The value of ε_r is **not constant** — it varies with: - Frequency - Temperature - Moisture absorption - For composite materials: relative proportions of components

2.2 FR4 Composition and Equations

(chap5.pdf, p.29–30)

FR4 = **epoxy matrix** reinforced by **woven glass cloth** → dielectric properties can **differ substantially** sample-to-sample.

First-order approximation (volume-weighted):

$$\varepsilon_r = \varepsilon_{\text{rsn}} V_{\text{rsn}} + \varepsilon_{\text{gls}} V_{\text{gls}}$$

(chap5.pdf, p.29)

where subscripts rsn = resin, gls = glass; V = volume fraction.

$$V_{\text{rsn}} \approx 1 - \frac{H_{\text{gls}}}{H}, \qquad V_{\text{gls}} = \frac{H_{\text{gls}}}{H}$$

(chap5.pdf, p.29)

Empirical equation from measurement:

$$\varepsilon_r(V_{\text{rsn}}, F) = 6.32 - [2.17 + 0.168 \log_{10} F(\text{kHz})] V_{\text{rsn}}$$

(chap5.pdf, p.29)

The glass cloth reinforcement has **no dielectric constant variation** with frequency in this range → epoxy resin (Vrsn) dominates.

> **Figure description (p.30):** Graph of relative dielectric constant ε_r vs. frequency (0.01 MHz to 1000 MHz) for FR4. The curve starts at ~4.7 at low frequencies and decreases monotonically to ~3.9 at 1000 MHz. The decrease is gradual and roughly logarithmic.

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3. Serpentine Traces

3.1 Why Serpentine Traces Are Used

(chap5.pdf, p.31)

Serpentine traces are used when: - Board **aspect ratios** and **real estate limitations** force routing constraints - Design specifications require **length equalization** (all traces same electrical length) but PCB area is limited

Key problem: Because of coupling between adjacent parallel sections of the serpentine, part of the signal propagates through **crosstalk coupling** (Path 2 = perpendicular shortcut through inter-segment coupling) rather than along the physical trace (Path 1). This **early arrival** reduces the effectiveness of length matching.

> **Figure description (p.31):** Diagram of a serpentine trace with labels: Driver, sections labeled L_p (parallel section length), spacing S, and Receiver. Two paths shown: Path 1 (along the physical trace) and Path 2 (through crosstalk, perpendicular shortcut across the parallel sections). Arrow and text note: "Signal will propagate on the trace AND through the crosstalk perpendicular to the coupled sections."

3.2 Effects on Signal Integrity and Timing

(chap5.pdf, p.32)

- **Duration of ledges** (early-arrival signal steps at the receiver) is proportional to the physical length of the coupled sections L_P
- **Voltage magnitude of ledges** depends on the spacing S between parallel sections

  Figure description (p.32): Top: Serpentine topology with 1-inch parallel sections and spacing S, cross-section showing trace width 5 mils, spacing S, $\varepsilon_r = 4.2$. Bottom: Two waveform plots at receiver. Left (edge rate = 30 ps): shows early arrival "ledge" before main step, for $S = 5$ mils and $S = 15$ mils vs. no serpentine. Right (edge rate = 250 ps): similar effect less visible. The earlier, smaller ledge is circled as "Early arrival."

3.3 Serpentine Trace Rules of Thumb

(chap5.pdf, p.33)

1. Make **minimum spacing** S between parallel sections at least **3H to 4H** (where H = height of signal conductor above ground plane) $\rightarrow$ minimizes coupling
2. **Minimize** L_P (length of serpentined sections) as much as possible $\rightarrow$ reduces total coupling magnitude
3. **Embedded microstrips and striplines** exhibit fewer serpentine effects than surface microstrip lines
4. **Do not serpentine clock traces**

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4. Intersymbol Interference (ISI)

4.1 Definition

(chap5.pdf, p.34)

ISI: When reflections, crosstalk, or any other source has **not settled completely** before the next bit transition, the signal launched onto the line will be affected by residual noise from previous bits.

- ISI is especially severe when the **period is smaller than** $2\times$ **the delay** of the transmission line
- To fully capture ISI effects: simulate with **long pseudorandom bit patterns** (PRBS)

4.2 Effect of ISI on Timing

(chap5.pdf, p.35)

ISI causes a **timing difference** between: - The ideal waveform (starting transition with no noise) - The noisy waveform (starting transition with unsettled noise on the bus)

> **Figure description (p.35):** Voltage vs. time plot showing two rising edges crossing the receiver switching threshold. The ideal edge (solid line) crosses at time t_1; the noisy edge (dashed line, starting from a different initial voltage due to ISI) crosses the threshold at a different time. The horizontal gap is labeled "Timing difference due to ISI." The noisy edge starts from "Different starting point due to ISI."

4.3 Alternative Simulation Method

(chap5.pdf, p.36)

To capture most timing impacts without full PRBS (which is expensive):

Simulate at three period multiples: - 1× fastest bus period → pattern: `010101010101010` - 2× fastest bus period → pattern: `001100110011001` - 3× fastest bus period → pattern: `000111000111000`

This is much faster than long PRBS while covering the key ISI scenarios.

4.4 Effect of ISI on Signal Integrity

(chap5.pdf, p.37)

Example: source impedance = 30 Ω, Z_0 = 65 Ω (overdriven case). ISI causes the waveform at the probe point to have different voltage levels depending on the previous bit pattern, causing: - Variable overshoot magnitudes - Variable settling levels - Eye closure

> **Figure description (p.37):** Schematic shows voltage source V with 30-ohm series resistor into Z_0 = 65-ohm TX line with probe at far end. Voltage vs. time waveform shows two overlaid bit patterns: 200 MHz (slower, larger swings) and 400 MHz (faster). The 400-MHz waveform shows ISI with varying peak levels and significant undershoots, while the ideal 400 MHz waveform (labeled) is shown for comparison.

4.5 ISI Rules of Thumb

(chap5.pdf, p.38)

1. **Minimize reflections** by avoiding impedance discontinuities; minimize stub lengths and large parasitics (packages, sockets, connectors)
2. **Keep interconnects short** as possible
3. **Avoid tightly coupled serpentine traces**
4. **Avoid line lengths** that cause signal integrity problems (ringback, ledges, overshoot) to coincide with the bus transition window
5. **Minimize crosstalk effects**

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5. Effects of 90° Bends

5.1 Capacitive Effect of a 90° Bend

(chap5.pdf, p.39)

A 90° bend in a PCB trace creates **excess capacitance** (more copper area than a straight trace at the corner). For typical 50–65 Ω line widths:

$$C_{90\check{r}\text{ bend}} \approx C_{11} w$$

(chap5.pdf, p.39)

where C_{11} = self-capacitance per unit length, w = line width.

The excess area (compared to straight trace) is approximately:

$$\text{Excess Area} = \frac{1}{4} - \frac{\pi}{16} \approx 0.054 \text{ squares}$$

(chap5.pdf, p.39)

> **Figure description (p.39):** Left: Top-view of a 90° corner trace with dimensions L_1, L_2, and width w. The corner area is highlighted. Right: Zoomed view of the corner showing the square corner (w-corner) vs. a 45° chamfer (w-chamfer), with the formula for excess area = 0.054 squares.

5.2 Ways to Reduce 90° Bend Capacitance

(chap5.pdf, p.40)

Three methods:

1. **Rounding**: Round the corners to maintain constant trace width around the bend → virtually eliminates the effect; but round corners cause problems with many layout tools
2. **Chamfering**: Cut the corner at 45° → reduces the excess area by removing the outermost triangle
3. **45° bends**: Completely replace 90° bends with 45° bends → eliminates the problem at the source

   > **Figure description (p.40):** Diagram comparing a 90° corner (w_corner) vs. a 45° chamfer ($w_chamfer$), showing how chamfering reduces the excess corner area.

5.3 Two Bends — Different Current Paths

(chap5.pdf, p.41)

In a trace with two 90° bends (U-shaped): - **High-frequency currents** take the shorter diagonal path (hugging the inner corner) → signals **arrive early** - **Low-frequency currents** take the longer outer path

> **Figure description (p.41):** U-shaped trace with two 90° bends. The outer physical path is shown in solid lines. A dashed diagonal arrow labeled "High frequency currents path" shows the shorter path that high-frequency signals can take through the inner corner area, arriving at the destination earlier than expected from the physical trace length.

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6. Effect of Topology

6.1 Balanced T Topology

(chap5.pdf, p.42)

A T-topology with **equal leg lengths** $L_1 = L_2$ and equal impedances Z_{o2} for both branches:

Matched case ($Z_{o2} = Z_{o1}$): clean step response to final value
Mismatched case ($Z_{o2} = 2Z_{o1}$): initial overshoot, then settles to correct final value (faster settling due to impedance step up)

Figure description (p.42): Circuit: source V_s (0–2V) with $R_s = Z_o \rightarrow$ main trunk line $Z_0 \rightarrow$ T-junction $\rightarrow$ two equal legs Z_{o2}, $L_1 = L_2$ to Receiver 1 and Receiver 2. Voltage at receiver vs. time (0–10 ns): $Z_{o2} = Z_{o1}$ shows smooth step to 2V; $Z_{o2} = 2Z_{o1}$ shows a larger initial step then settles at 2V.

6.2 Unbalanced T Topology — Matched Load

(chap5.pdf, p.43–44)

When leg lengths are **unequal** ($L_1 < L_2$), even with matched loads at each receiver, multiple reflections bounce between the junction and the receivers causing severe signal integrity degradation.

Lattice diagram analysis for unbalanced T:

Initial voltage:

$$V_{\text{initial}} = 2V \frac{Z_o}{Z_o + R_s} = 1\text{V}$$

(chap5.pdf, p.43)

Reflection coefficient at each branch (branch impedance = $Z_o/2$ = parallel of two Z_o legs):

$$\rho_2 = \rho_3 = \frac{Z_o/2 - Z_o}{Z_o/2 + Z_o} = -\frac{1}{3}$$

(chap5.pdf, p.43)

Transmission to each branch:

$$T_2 = T_3 = 1 + \rho = \frac{2}{3}$$

(chap5.pdf, p.43)

Successive voltage levels at Receiver 1 (shorter leg L_1):

Event	Voltage
A: initial arrival	$2/3 + 2/3 = 4/3 = 1.33$ V
B: first reflected arrival	$4/3 + 2/3 - 2/9 - 2/9 = 8/9 = 0.888$ V
C: second reflected arrival	$\cdots = 16/9 = 1.77$ V

At Receiver 2 (longer leg): - $A' = 4/3$, $B' = 20/9 = 2.22$ V, then ring around V=2

Result: Signal integrity **deteriorates dramatically** with multiple oscillations.

Figure description (p.44): Lattice diagram with 4 vertical lines (A=Receiver 1, B=Junction, C=Junction, D=Receiver 2) showing bouncing waves. Voltage vs. time plot at receiver showing oscillating waveform peaking at $20/9 = 2.22$ V initially, then oscillating around 2 V and never cleanly settling. Source = 0-2V, $R_s = Z_o$, unequal legs $L_1 < L_2$.

6.3 Lessons Learned from Topology Analysis

(chap5.pdf, p.44)

1. **Symmetry topology is critical**: leg lengths AND loading at each branch must be equal
2. **Minimize discontinuities** at topology junctions (impedance mismatches at the T-point)

Key Equations Summary

Quantity	Formula	Reference
DC resistance	$R = \rho L/(Wt)$	*(chap5.pdf, p.4)*
Skin depth	$\delta = \sqrt{\rho/(\pi F \mu)}$	*(chap5.pdf, p.6)*
AC signal resistance	$R_{\text{ac sig}} \approx \sqrt{\rho\pi\mu F}/W$	*(chap5.pdf, p.9)*
Ground return AC resistance	$R_{\text{ac gnd}} \approx \sqrt{\rho\pi\mu F}/(6H)$	*(chap5.pdf, p.11)*
Total AC microstrip resistance	$R_{\text{ac}} \approx \sqrt{\rho\pi\mu F}(1/W+1/6H)$	*(chap5.pdf, p.12)*
AC stripline resistance	$R_{\text{ac strip}} = R_{H1} \| R_{H2}$ (parallel)	*(chap5.pdf, p.13)*
Surface resistance	$R_s = \sqrt{\rho\pi\mu}$ (ohms$\cdot\sqrt{\text{s}}$/square)	*(chap5.pdf, p.16)*
AC resistance from R_s	$R_{\text{ac}} = R_s\sqrt{F}$	*(chap5.pdf, p.15)*
Square wave Fourier series	$f(x) = \frac{2}{\pi}\sum_{n=1,3,5}\frac{1}{n}\sin(2\pi nFx)$	*(chap5.pdf, p.17)*
Attenuation factor	$\alpha = R_2/(R_{\text{ac}} + R_2)$	*(chap5.pdf, p.18)*
Complex dielectric constant	$\varepsilon = \varepsilon' - j\varepsilon''$	*(chap5.pdf, p.20)*
Loss tangent	$\tan\delta_d = \varepsilon''/\varepsilon'$	*(chap5.pdf, p.21)*
Dielectric conductance	$G = (\varepsilon''/\varepsilon')(2\pi F C_{11})$	*(chap5.pdf, p.21)*
FR4 dielectric constant	$\varepsilon_r(V_{\text{rsn}}, F) = 6.32 - [2.17 + 0.168\log_{10} F(\text{kHz})]V_{\text{rsn}}$	*(chap5.pdf, p.29)*
90° bend capacitance	$C_{90\check{r}} \approx C_{11}w$	*(chap5.pdf, p.39)*
Unbalanced T: ρ at branch	$\rho = (Z_o/2 - Z_o)/(Z_o/2 + Z_o) = -1/3$	*(chap5.pdf, p.43)*
Unbalanced T: transmission	$T = 1 + \rho = 2/3$	*(chap5.pdf, p.43)*